

Gerard Consuelo

San Jose, CA, 95112 | gcon.work@gmail.com | linkedin.com/in/gerard-consuelo | gerardconsuelo.com

EDUCATION

San Jose State University

Bachelor of Science in Computer Science

- GPA: 3.93, 3x President's Scholar Award.

San Jose, CA

Graduation: Dec 2027

EXPERIENCE

Software and Computer Engineering Society (SCE)

Aug. 2024 – Present

Software Engineering Intern

San Jose, CA

- Collaborated to build a non-intrusive occupancy system scanning bluetooth from nearby phones.
- Architected a system with an ESP32 BLE scanner, MQTT broker, and REST API updating to a public website.
- Processed over 100k+ addresses per day with MAC address bit-mask filtering ensuring accurate headcounts.
- Forked and rewrote ESPresence firmware to support WaveShare PoE-powered Ethernet ESP32 boards.

SJSU College of Engineering

May 2025 – Aug. 2025

Software Engineer Intern

San Jose, CA

- Architected a real-time Python and SQLite backend service for monitoring print jobs for over 2,000+ people.
- Ensured 99% uptime for 500+ jobs by automating Pytest unit tests within GitHub Actions CI/CD pipelines.
- Reduced memory usage by 65% via Docker consolidation to streamline deployment on a bare-metal Linux server.

Lead Instructor

Jul. 2025

theCoderSchool

Folsom, CA

- Led week-long camp of 15 students with 1 TA, achieving 100% project completion and zero parent complaints.
- Designed 8-hour LuaU curriculum with 5 games, adopted as standard training material for future camps.
- Boosted daily quiz scores 56% to 84%, guiding students to produce presentations and 10+ multiplayer games.

PROJECTS

AgentClipz | *Python, FFmpeg, Streamlink, Reka AI, Modulate API, Claude Code, Context7 MCP*

- Awarded **Best Use of Modulate** out of 80+ teams for a live-stream clip generator using sentiment analysis.
- Processed 4-hour live streams via Streamlink and FFmpeg, using chat spikes and Modulate to audit clips.
- Published Reka AI-captioned Reels to Instagram 2 minutes after the event occurred, securing 1.5K views.

StudyGuard | *Python, FastAPI, OpenCV, OpenAI Whisper, NVIDIA Jetson Orin Nano*

- Awarded **2nd Overall** (15 teams) for a study robot that reloads dynamic DNS to block distractions.
- Deployed OpenCV on a Jetson Orin Nano to classify current study states, streaming data to a dashboard.
- Used a Brev-hosted Llama VL model and Whisper STT to unblock study sites on demand after analysis.

SOW-agent | *Python, AWS Bedrock, GitHub Actions, FastAPI*

- Architected a 5-agent AWS Bedrock system automating SOW compliance, generating implementation code.
- Developed a self-healing QA loop with Amazon Nova, auto-retrying failed agent code generations up to 3 times.
- Built a Next.js dashboard to execute agents against GitHub repositories, auto-pushing compliant code to PRs.

Schedule-AI | *Python, FastAPI, React, Turso SQLite, SQLAlchemy, PlaywrightMCP, Groq*

- Engineered an automated degree planner, scraping 5.3K courses and 200 majors via Playwright and Groq LLMs.
- Integrated a RAG pipeline using LangChain, Groq, ChromaDB, and embedding models for data retrieval.
- Built a FastAPI and Turso SQLite backend to generate dynamic 4-year degree roadmaps using junction tables.

TECHNICAL SKILLS

Languages: Python, Go, C, Java, SQL, LuaU, JavaScript, HTML/CSS

Frameworks/Libraries: FastAPI, LangChain, PyTorch, Transformers, HuggingFace, ChromaDB, OpenAI API, OpenCV, SQLAlchemy, React/Next.js

Developer Tools: Docker, Linux/Unix, Git/GitHub Actions, Pytest, Claude Code, Gemini, MCP, Groq, AWS Bedrock